

B V MANOJ

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PROFESSIONAL SUMMARY

AI, Machine Learning & Data Science undergraduate (Cumulative GPA 9.2/10) building production-grade Generative AI and Large Language Model (LLM) applications, RAG systems, and RL environments. 5 deployed projects spanning multi-LLM orchestration, video AI tooling, component risk analysis, and autonomous news intelligence. Proficient in Python, TypeScript, PyTorch, TensorFlow, FastAPI, Next.js, FAISS, Ollama, HuggingFace Transformers, and AWS.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS

AI/ML: PyTorch, TensorFlow, HuggingFace Transformers, LLMs (OpenAI, Anthropic, Ollama), RAG (FAISS, LangChain), Prompt Engineering, Generative AI, NLP, Deep Learning, Computer Vision (OpenCV), Reinforcement Learning (OpenAI Gym), Scikit-learn, Pandas, NumPy

LLM/APIs: Multi-LLM fallback chains, Genkit, Firebase AI, Telegram Bot API, RESTful API design

Data: SQLite, PDF/DOCX parsing, Web scraping, Data pipelines, ETL

Cloud/DevOps: AWS (EC2), Google Cloud Run, Vercel, Firebase, Git/GitHub, MLOps (CI/CD, versioning), Docker

Tools: VS Code, Postman, Jupyter, Figma (basics), Linux CLI, AI Agents

Soft Skills: Problem solving, Technical communication, Rapid prototyping, Self-directed learning

Spoken Languages: English (Fluent), Telugu (Native), Hindi (Conversational)

EXPERIENCE

videoreverse

Python | github.com/Venkata-Manoj/videoreverse

Tool to deconstruct videos into production-ready prompts for video AI models.

- Engineered video frame extraction pipeline using OpenCV reducing prompt engineering time by 70%
- Implemented scene detection and keyframe selection algorithms improving AI model output relevance by 45%
- Designed CLI interface with batch processing enabling 100+ video analysis per session

AI-News-Bot

Python, PyTorch, Multi-LLM, SQLite | github.com/Venkata-Manoj/AI-News-Bot

Autonomous news intelligence with 6-LLM fallback chain scraping 6 sources, delivering Telegram cards every 45 minutes.

- Architected 6-LLM fallback chain (OpenAI, Anthropic, Ollama) achieving 99.2% uptime over 30 days
- Built multi-source scraper processing 500+ articles/day with deduplication reducing noise by 60%
- Deployed scheduled Telegram bot delivering formatted cards to 200+ subscribers with 45-min cadence

WhatIF

TypeScript, Next.js, Firebase, Genkit | github.com/Venkata-Manoj/WhatIF

AI-powered UI component analyzer for React/Vue/HTML with instant risk identification and exportable PDF reports.

- Developed Genkit-powered analysis engine detecting 15+ risk categories (accessibility, performance, security) in <3s
- Integrated Firebase Auth + Firestore for user sessions scaling to 500+ analyses in first month
- Built PDF report generation with risk scoring reducing manual code review effort by 80%

Capstone-Forge

Python, FastAPI, FAISS, Ollama | github.com/Venkata-Manoj/Capstone-Forge

RAG-powered report generator ingesting PDFs/DOCX/images to produce institution-compliant capstone reports.

- Implemented FAISS vector database store with 10k+ document chunks achieving 92% retrieval accuracy on tests
- Built FastAPI backend with Ollama local LLM inference cutting cloud costs by 100% vs API-based approach
- Designed template engine generating formatted reports meeting 100% of institutional compliance checks

EDUCATION

B.Tech Artificial Intelligence & Data Science

Jun 2024 - Present

Saveetha School of Engineering (SIMATS) | Cumulative GPA (CPI): 9.2/10

HSC (12th)

Jun 2022 - May 2024

SR Junior College | 95.6%

SSC (10th)

May 2022

Z.P. High School | 85%

CERTIFICATIONS

- DataCamp: Introduction to LLMs in Python (2024)
- Anthropic: Prompt Engineering Fundamentals (2024)
- DataCamp: Introduction to RAG with LangChain (2024)
- DataCamp: Building Multi-Step LLM Chains (2024)

AWARDS & ACHIEVEMENTS

- 1st Prize - Vel Tech Hackathon 2024 (AI-powered healthcare solution)
- Smart India Hackathon 2025 - National finalist (Top 50/15000+ teams)
- SIMATS Hackathon Club - Core member organizing 5+ events for 200+ participants